Example

Minimize $0.5x^2 + 3x + 4y$

Minimize $0.5x^2 + 0y^2 + 0xy + 3x + 4y$

Minimize $\frac{1}{2}[x^2 + 0y^2 + 0xy] + 3x + 4y$

$[x^T P x] + q^T x$

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad q = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Subject to

$$\begin{array}{l} x \geq 0 \\ y \geq 0 \\ x + 3y \geq 15 \\ 2x + 5y \leq 100 \\ 3x + 4y \leq 80 \end{array} \Rightarrow \begin{array}{l} -x \leq 0 \\ -y \leq 0 \\ -x - 3y \leq -15 \\ 2x + 5y \leq 100 \\ 3x + 4y \leq 80 \end{array}$$

$$Gx \leq h$$

$$G = \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ -1 & -3 \\ 2 & 5 \\ 3 & 4 \end{bmatrix} \quad h = \begin{bmatrix} 0 \\ 0 \\ -15 \\ 100 \\ 80 \end{bmatrix}$$

x=0, y=5

$$\begin{array}{l} x + 3y = 15 \Rightarrow \geq 15 \\ 2x + 5y = 25 \Rightarrow \leq 100 \\ 3x + 4y = 20 \Rightarrow \leq 80 \\ x \geq 0 \\ y \geq 0 \end{array}$$

Minimum value:

$0.5x^2 + 3x + 4y = 20$